

🛡 SpoofVane — Complete Feature Catalogue

Everything in the product: 76 backend capability modules, 7 Bright Data products, the platform & governance layer, and 21 console screens.

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76

capability
modules

7

Bright Data
products

21

console screens

350

tests passing

15k+

lines of code

Capability modules — 8 groups, 76 features

● Discovery 10 modules

How SpoofVane finds impersonation

A1

SERP scanner

Bright Data SERP API
(Google/Bing/Brave/DuckDuckGo, brd_json=1);
paginates & dedupes.

A2

Cert transparency

Tails CT logs (Calidog certstream) + Web Scraper
API fallback for new lookalike certs.

A3

DNS brute

Permutation + bitsquatting + homoglyph + IDN;
resolves via residential proxies.

A4

Registrar feed

Bright Data Datasets WHOIS feed; flags newly-
registered & risky-country domains.

A5

Phishing feeds

OpenPhish + PhishTank + URLhaus + UrlAbuse
ingest, normalised to one IOC schema.

A6

Social impersonation

X / IG / TikTok / LinkedIn / YouTube via SERP +
Web Unlocker.

A7

A8

URL shortener

Expands bit.ly/t.co/lnkd.in chains, recording every redirect hop to the terminal URL.

Ad network

Google/Meta/TikTok ad libraries via Web Scraper API; captures creative+advertiser+landing.

A9

Mobile app store

iOS App Store + Google Play scraping for fake apps; matches name + icon pHash.

A10

Crawl seed

Customer seed lists + dark-web mention adapters (HudsonRock/SpyCloud pattern), idempotent.

● Inspection 8 modules

How each suspect is captured & analysed

B1

Browser inspector

Bright Data Scraping Browser over CDP, country-pinned; DOM+screenshot+HAR+final URL.

B2

Cloaking detector

Renders ≥ 2 geos and diffs DOM+pHash+content to catch geo-cloaking.

B3

DOM extractor

Classifies forms — login / payment / iframe — from the real parsed DOM.

B4

TLS inspector

Cert chain + JA3/JA4 fingerprint + SCT; flags self-signed.

B5

HAR collector

Full HAR with per-resource SHA-256, waterfall timing, external-origin tagging.

B6

pHash extractor

Perceptual + difference hashes on screenshots and logo crops.

B7

WHOIS enricher

WHOIS/RDAP + hosting ASN + rDNS; flags bulletproof hosts.

B8

Ad creative capture

Screenshot + OCR of an ad creative paired with its landing page.

● Scoring 10 modules

How similarity & risk are quantified

C1

C2

Composite scorer

Platt-calibrated probability (not a raw sum) with per-signal contribution breakdown.

URL risk scorer

Entropy + suspicious tokens + TLD + punycode + IP + age → risk with contributions.

C3

DOM scorer

Structural tree-edit distance vs known-good login pages.

C4

pHash scorer

Hamming distance over perceptual hashes vs brand assets.

C5

Logo scorer

CLIP/MobileCLIP logo detection (feature-flagged, never 0% from missing weights).

C6

Kit fingerprint

13 kit signatures (EvilProxy, Tycoon-2FA, Greatness, Mamba2FA...) via DOM + JS-bundle hash.

C7

Family classifier

ML classifier mapping kit features → threat family.

C8

Cluster score

Union-find threat-network clustering by shared ASN/registrar/kit/favicon/JA3.

C9

Voiceprint score

192-dim speaker embedding; cosine match against an enrolled executive voiceprint.

C10

Deepfake score

Ensemble: face embedding + lip-sync + C2PA provenance → deepfake probability.

● Verdict 8 modules

How the final call is made

D1

Claude verdict

Claude structured-output verdict with prompt-guard and recorded rationale.

D2

GPT verdict

GPT verdict for the ensemble, same schema.

D3

Gemini verdict

Gemini verdict for the ensemble, same schema.

D4

SLM triage

Small open-weight model — the cheap path for low-severity triage.

D5

Verdict merger

Disagreement-aware merge; discounts confidence and escalates to a human on dissent.

D6

Multimodal verdict

Fuses text + image + audio for deepfake verdicts.

D7

MITRE enricher

Maps every alert to real ATT&CK technique IDs + D3FEND countermeasures.

D8

Cache + active learning

Verdict cache keyed on evidence hash; analyst overrides retune calibration.

● Agents 10 modules

Autonomous, governed actions

E1

Takedown agent

Researches → drafts → HITL gate → submits registrar packets.

E2

Victim-id agent

Identifies victims from session telemetry (consent + lawful-basis gated).

E3

Cred-poison agent

Submits unique canary credentials per URL and tracks dark-web reuse.

E4

Synth-pages agent

Deploys honeypot pages back at the attacker in an isolated namespace.

E5

Cluster agent

Maintains the threat-network graph incrementally.

E6

Learning agent

Wires analyst overrides into the C1 calibration loop.

E7

Kill switch

Admin hard-halt for every running agent (global or per-tenant) + audit.

E8

Governance

Scores blast-radius; enforces budget/cost/risk envelope + HITL above threshold.

E9

Agent audit

E10

SLM triage agent

Hash-chained, tamper-evident entry for every agent action.

Agentic wrapper around the D4 cheap-path model.

● Delivery 8 modules

Takedown & integrations out

F1

Cloudflare takedown

Real Cloudflare abuse-form POST behind HITL; returns a reference id.

F2

GoDaddy takedown

GoDaddy abuse API submission.

F3

Namecheap takedown

Namecheap abuse API submission.

F4

Hosting abuse

AWS/GCP/Hetzner/OVH/Hostinger/Cloudflare per-host abuse channel routing.

F5

SIEM/SOAR sinks

Splunk + Sentinel + ServiceNow + PagerDuty + Cortex XSOAR, with retry/backoff.

F6

TAXII / STIX

Schema-valid STIX 2.1 bundle export over TAXII 2.1.

F7

Own MCP server

SpoofVane's MCP server — analyst-in-Claude-Desktop with 14+ tools.

F8

BD MCP client

Calls Bright Data's MCP server for ad-hoc BD-routed rendering.

● Platform 12 modules

Enterprise foundations

G1

Identity

OIDC + SAML + JWT + signed cookies + MFA + SCIM 2.0.

G2

RBAC

6 roles with explicit segregation of duties.

G3

Audit logger

G4

HITL review

Mandatory review queue; reviewer ≠ analyst.

SHA-256/Blake2b hash-chained audit, Ed25519-signed.

G5

Notifications

Email + Slack + Teams + WhatsApp + push + voice (Twilio/Plivo/Telnyx).

G6

Reports

Executive PDF + board pack + auditor pack, PKCS#7/PyHanko-signed.

G7

Rate limiter

Per-tenant token bucket.

G8

Idempotency

Idempotency-key middleware to make retried side-effects safe.

G9

Cost tracker

Per-tenant Bright Data + LLM spend telemetry vs envelope.

G10

Deepfake compliance

Chain-of-custody pack for legal discovery.

G11

BYOK

Customer-managed keys (AWS KMS / Azure KV / GCP KMS) via envelope encryption.

G12

Data residency

EU/US/APAC/IN routing + provable non-egress.

● AI Surfaces 10 modules

Analyst-facing intelligence

H1

Analyst copilot

Conversational verdict / triage assistant.

H2

Audit NL search

Natural-language search over the audit log → structured filters.

H3

Brand wizard

Conversational brand onboarding (keywords, TLDs, regions).

H4

Deepfake verdict UI

Analyst-facing deepfake verdict surface.

H5

H6

Exec attack surface

Executive likeness + voice surface monitor (sensitive, gated).

Family reranker

Re-ranks family classification with analyst feedback.

H7

Intel narrator

Auto-generates the human-readable alert narrative.

H8

Kit explainer

RAG over the phishing-kit knowledge base (prompt-cached).

H9

Takedown drafter

Drafts the registrar packet for analyst review.

H10

TTP proposer

Suggests MITRE TTPs for the alert.

Bright Data — 7 sponsor products

BD 1

Scraping Browser (CDP)

Country-pinned residential rendering of suspect pages

used by B1, B8, F1

BD 2

Residential Proxies

Geo-distributed HTTP + DNS from real residential exits

used by A3, A6, B1

BD 3

SERP API

brd_json=1 structured search across 4 engines

used by A1

BD 4

Web Unlocker

Bypasses JS challenges / 403s on protected targets

used by A6, A9, B1

BD 5

Web Scraper API

Structured extraction of ads & fallback pages

used by A2, A8, B8

BD 6

Datasets / Marketplace

WHOIS dataset feed for registration intelligence

used by A4, B7

BD 7

Bright Data MCP Server

Ad-hoc BD-routed rendering from inside the copilot

Platform & governance highlights



Replay mode

Every Bright Data path runs credential-free against input-dependent fixtures (reproducible build).



Differential depth probe

14 module probes assert output varies with input — stubs can't pass.



Two-person rule

Egress agents (cred-poison, synth-pages) need a second authoriser + permitted region.



Tamper-evident chains

Audit, agent-audit and chain-of-custody each hash-chain and self-verify.



Cost envelope

Per-tier caps (free \$5 / pro \$100 / business \$1k / enterprise \$25k) with SLM-first routing.



CycloneDX SBOM

26-component software bill of materials generated from the real dependency tree.

Modules map to the review's §8 catalogue (A1–H10) and the canonical page registry. Every feature has real, tested, input-dependent code – see `docs/TRACEABILITY_MASTER.md` and `docs/DEPTH_LEDGER.md`.